

Econ 101A

Section 3 (Solutions)

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This section is about **optimization under a constraint**, the workhorse of the whole course. The consumer maximizes utility subject to a budget; the firm minimizes cost subject to producing a given output. The tool is the **Lagrangian**, and it connects directly to the two ideas from Section 2: the tangency condition is $MRS = \text{price ratio}$, and the multiplier is a shadow price you can read off with the envelope theorem.

1 The Constrained Problem and the Lagrangian

We want to solve

$$\max_{x,y} f(x,y) \quad \text{subject to} \quad g(x,y) = b,$$

where f is the objective and $g(x,y) = b$ is one equality constraint. Form the **Lagrangian**

$$\mathcal{L}(x,y,\lambda) = f(x,y) + \lambda(b - g(x,y)),$$

and treat λ (the **Lagrange multiplier**) as an extra variable. At an interior optimum the first-order conditions are

$$\mathcal{L}_x = f_x - \lambda g_x = 0, \quad \mathcal{L}_y = f_y - \lambda g_y = 0, \quad \mathcal{L}_\lambda = b - g(x,y) = 0.$$

The last equation just returns the constraint.

The tangency condition. Divide the first two FOCs to eliminate λ :

$$\frac{f_x}{f_y} = \frac{g_x}{g_y}.$$

In the consumer's problem $f = u$ and $g(x,y) = p_x x + p_y y$ (so $g_x = p_x$, $g_y = p_y$), and this becomes

$$MRS = \frac{u_x}{u_y} = \frac{p_x}{p_y}.$$

The best affordable bundle is where the indifference curve is *tangent* to the budget line: the rate at which you are *willing* to trade x for y equals the rate at which the *market* lets you.

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The general statement (gradients are parallel). For $\max_{x \in \mathbb{R}^n} f(x)$ subject to $h(x) = 0$, if $\nabla h(x^*) \neq 0$ then there is a multiplier λ with

$$\nabla f(x^*) = \lambda \nabla h(x^*).$$

At the optimum the gradient of the objective is parallel to the gradient of the constraint. That is the same tangency statement, written with gradients: you cannot improve f without pushing off the constraint surface.

2 Second-Order Conditions: the Bordered Hessian

The FOCs only find *stationary* points. To confirm a maximum you check a second-order condition, and with a constraint the right object is the **bordered Hessian**. For two choice variables and one constraint $g(x, y) = b$, form

$$\bar{H} = \begin{bmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix}, \quad \mathcal{L}_{xx} = f_{xx} - \lambda g_{xx}, \text{ etc.,}$$

with everything evaluated at the candidate (x^*, y^*, λ^*) . Then the stationary point is a

local constrained maximum if $\det \bar{H} > 0$, **local constrained minimum** if $\det \bar{H} < 0$.

The border row/column encodes the directions you are allowed to move along the constraint; the determinant checks the curvature of f restricted to those directions. In practice you rarely need this by hand: if the objective is concave and the constraint linear (as with a strictly quasi-concave utility and a budget line), the FOCs already deliver the global maximum. Keep the bordered Hessian for the cases where concavity is not obvious.

3 What the Multiplier λ Means

λ is the **shadow price** of the constraint: the rate at which the optimized objective rises when you relax the constraint by one unit. Let $V(b) = \max f$ subject to $g = b$ be the value function. By the envelope theorem (Section 2),

$$\frac{dV}{db} = \lambda.$$

In the consumer's problem, define the **indirect utility** $v(p_x, p_y, m) = u(x^*, y^*)$, the best utility attainable at those prices and income. Then

$$\lambda = \frac{\partial v}{\partial m} :$$

the multiplier is literally the **marginal utility of income**, how much extra utility one more dollar buys.

One caveat. The tangency FOCs describe an *interior* optimum with the constraint binding. With perfect substitutes, or when a non-negativity constraint bites, the optimum can be a **corner**; always sanity-check whether an interior tangency actually exists.

4 Exercises

Exercise 1 (constrained optimization, pure math)

Solve

$$\max_{x,y} f(x,y) = x^2y \quad \text{subject to} \quad x + y = 12, \quad x, y \geq 0.$$

- Write the Lagrangian and the first-order conditions.
- Find the optimal (x^*, y^*) , the maximum value, and λ .
- Explain briefly why the stationary point you found is the maximum (rather than a minimum).
- Confirm the second-order condition at (x^*, y^*) using the bordered Hessian.

Solution. (a) $\mathcal{L} = x^2y + \lambda(12 - x - y)$. FOCs: $\mathcal{L}_x = 2xy - \lambda = 0$, $\mathcal{L}_y = x^2 - \lambda = 0$, $\mathcal{L}_\lambda = 12 - x - y = 0$.

(b) Setting the first two equal (both equal λ): $2xy = x^2 \Rightarrow x(2y - x) = 0$. The relevant root is $x = 2y$ (the root $x = 0$ gives $f = 0$). With $x + y = 12$: $2y + y = 12 \Rightarrow y^* = 4$, $x^* = 8$. Then $f^* = x^2y = 64 \cdot 4 = 256$ and $\lambda = x^2 = 64$.

(c) On the feasible segment the objective is 0 at both endpoints ($x = 0$ or $y = 0$) and strictly positive in between, so the interior stationary point $(8, 4)$ with $f = 256 > 0$ is the maximum. (Equivalently, substitute $y = 12 - x$ to get $h(x) = x^2(12 - x)$; $h'(x) = 3x(8 - x) = 0$ at $x = 8$ and $h''(8) = -24 < 0$.)

(d) Here $g(x, y) = x + y$ so $g_x = g_y = 1$, and $\mathcal{L}_{xx} = f_{xx} = 2y$, $\mathcal{L}_{xy} = f_{xy} = 2x$, $\mathcal{L}_{yy} = f_{yy} = 0$. Thus

$$\bar{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2y & 2x \\ 1 & 2x & 0 \end{bmatrix}, \quad \det \bar{H} = 4x - 2y.$$

At $(8, 4)$, $\det \bar{H} = 4(8) - 2(4) = 24 > 0$, which confirms a constrained *maximum*.

Exercise 2 (utility maximization, applied)

A consumer chooses (x, y) to maximize $u(x, y) = \alpha \ln x + \beta \ln y$ (with $\alpha, \beta > 0$) subject to the budget constraint $p_x x + p_y y = m$, where $p_x, p_y, m > 0$.

- Write the Lagrangian and the first-order conditions.
- Derive the tangency condition and solve for the Marshallian demands $x^*(p_x, p_y, m)$ and $y^*(p_x, p_y, m)$.
- What fraction of income is spent on each good? Interpret λ .
- Now replace preferences with $u(x, y) = 2x + y$ (perfect substitutes). Explain why the tangency approach can fail, and give the optimal bundle as a function of prices.

Solution. (a) $\mathcal{L} = \alpha \ln x + \beta \ln y + \lambda(m - p_x x - p_y y)$. FOCs: $\frac{\alpha}{x} - \lambda p_x = 0$, $\frac{\beta}{y} - \lambda p_y = 0$, $p_x x + p_y y = m$.

(b) Dividing the first two: $\frac{\alpha/x}{\beta/y} = \frac{p_x}{p_y}$, i.e. $\text{MRS} = \frac{\alpha y}{\beta x} = \frac{p_x}{p_y}$ (tangency). From the FOCs, $p_x x = \frac{\alpha}{\lambda}$ and $p_y y = \frac{\beta}{\lambda}$; adding and using the budget, $m = \frac{\alpha + \beta}{\lambda}$, so $\lambda = \frac{\alpha + \beta}{m}$. Hence

$$x^* = \frac{\alpha}{\alpha + \beta} \frac{m}{p_x}, \quad y^* = \frac{\beta}{\alpha + \beta} \frac{m}{p_y}.$$

(c) Expenditure on x is $p_x x^* = \frac{\alpha}{\alpha + \beta} m$, a *constant* share $\frac{\alpha}{\alpha + \beta}$ of income (and $\frac{\beta}{\alpha + \beta}$ on y). Constant budget shares are the hallmark of Cobb–Douglas. The multiplier $\lambda = \frac{\alpha + \beta}{m}$ is the marginal utility of income: by the envelope theorem $\frac{\partial v}{\partial m} = \lambda$, so one extra dollar raises maximized utility by about λ .

(d) With $u = 2x + y$, $\text{MRS} = \frac{u_x}{u_y} = 2$ is constant. The tangency condition $\text{MRS} = p_x/p_y$ then holds only in the knife-edge case $p_x/p_y = 2$; otherwise the interior FOCs are inconsistent (they would demand $2 = p_x/p_y$), and the optimum is a **corner**. Compare willingness-to-pay to the price ratio: if $2 > \frac{p_x}{p_y}$ (i.e. $p_x < 2p_y$), spend everything on x : $x = \frac{m}{p_x}$, $y = 0$; if $2 < \frac{p_x}{p_y}$, spend everything on y : $y = \frac{m}{p_y}$, $x = 0$; if $p_x/p_y = 2$, any point on the budget line is optimal.